

Uncovering Temporal Framing in the News

Beyond chronology: how temporal language shapes interpretation.

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Beyond Chronology and Timelines

Two sentences. Same topic. Different rhetorical work.

FACTUAL CHRONOLOGICAL REPORT

Crime rose by 2% last year.

Statement locates a statistic on a timeline.

RHETORICAL TEMPORAL FRAMING

Once a booming hub, now a city in decline.

Contrasts 2 time periods to shape interpretation.

How NLP Traditionally Deals with Time

"Years after the election, the country is still haunted by its aftermath."

Traditional Temporal NLP

- Extracts temporal expressions
- Detects events
- Performs temporal reasoning

When did things happen?

Temporal Framing (Our Work)

- Why invoke 'years after'?
- What does 'still' suggest?
- How does time shape interpretation?

What is time doing rhetorically?

DEFINITION

From Chronology to Persuasion



Temporal framing

Rhetorical use of time-related language to structure meaning.



Task

Given a sentence, identify which temporal frames are active.



Output

One or more labels from an 8-frame taxonomy.

A Taxonomy of Temporal Frames

Eight different ways temporal language can be used rhetorically.



Primacy

Being first matters



Continuity

Persistence over time



Nostalgia

Valued past



Temporal Contrast

“Then” vs. “now”



Recency

Recent events matter



Skeptical

Doubt about the future



Urgency

Time pressure



Temporal Anchoring

Past event as a lens

EXAMPLE

Annotated Example

Violence Has Been Normalized

During the misnamed and mostly preposterous debate between Kamala Harris and Donald Trump, a moderator fact-checked Trump's claim that crime is up.

In contrast to Trump's claim, moderator David Muir said that the FBI reports that crime is down, a claim that likely struck every viewer as obviously wrong.

Shoplifting was not a way of life before lockdowns. Nostalgia Temporal Anchoring Temporal Contrast Most cities were not demographic minefields of danger around every corner. Nostalgia Temporal Contrast There was no such thing as a drugstore with nearly all products behind locked Plexiglas. Nostalgia Temporal Contrast

We weren't warned of spots in cities, even medium-sized ones, where carjacking was a real risk. Nostalgia Temporal Contrast

It is wildly obvious that high crime in the U.S. is endemic, with ever less respect for person and property. Continuity Primacy
As for the FBI's statistics, they're worth about as much as most data coming from federal agencies these days. Temporal Contrast

Temporal frames are **sentence-level** and **multi-label**.

DATASET

A Multilingual Corpus for Temporal Framing

Sentence-level, human-annotated, multi-label dataset from real news.

458

News Articles

English & German · 35+ outlets

238 · 220

EN · DE

Document counts

319 / 63 / 76

Train / Dev / Test

70% / 14 % / 16%

20,659

Sentences

Segmented from full article text

2,365

Framed Sentences

~11.5% prevalence

3,030

Annotations

~1.28 labels / framed sent.

Can Models Learn Temporal Framing?

We compare zero-shot with supervised fine-tuning across model families & scales.

Zero-shot

No task-specific training



GPT-5.2



LLaMA-3.1 (8B-70B)



Qwen3 (8B-235B)

Supervised fine-tuning

Trained on our corpus



XLM-R (270M)



Qwen3-8B



LLaMA-3.1-8B

Does scale beat supervision?

RESULTS

Supervision Beats Scale



Best Overall

Qwen3-8B (FT)

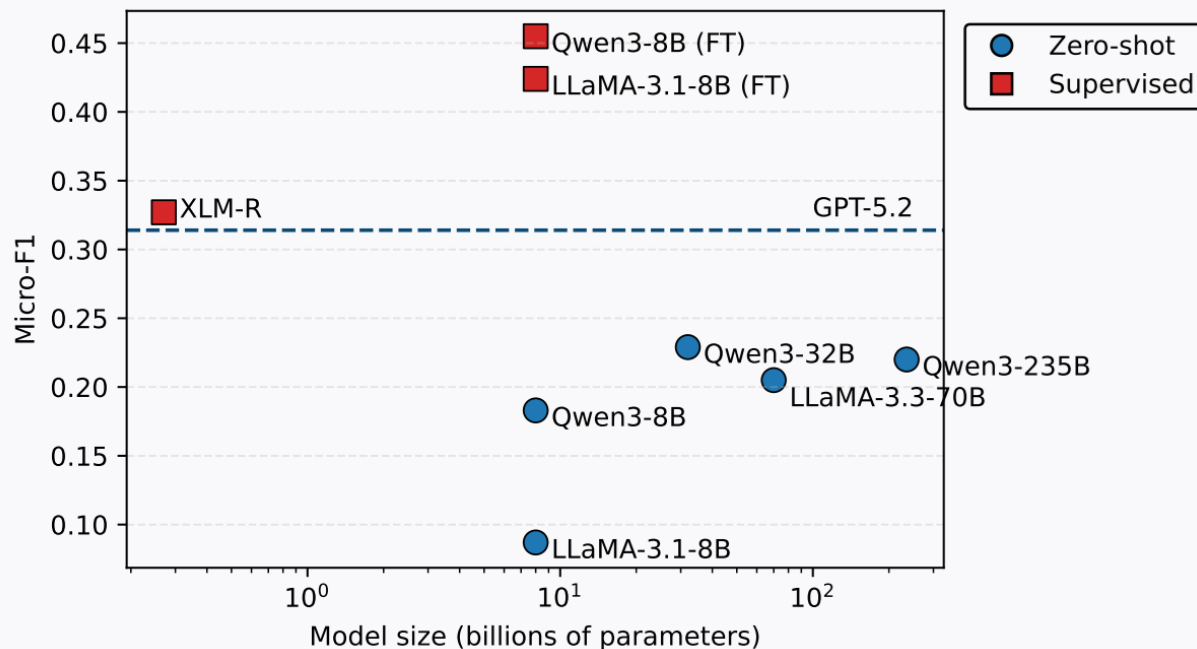
+27% over best zero-shot



Efficient Baseline

XLM-R (270M)

Beats **ALL** zero-shot models



Why Some Frames Are Harder

Surface cues help. Some frames exhibit more cues than others.



Easier: lexical cues

Stable patterns in the text



Primacy

record, first, ever,
never, history, since



Continuity

steady, increasingly,
repeatedly, streak



Harder: rhetorical intent

No reliable lexical signature



Anchoring

None



Nostalgia

None

What This Paper Contributes

- 1 Task** Temporal framing as a sentence-level, multi-label NLP task.
- 2 Taxonomy** Eight temporal frames for rhetorical use of time.
- 3 Resource** Human-annotated English-German news corpus.
- 4 Analysis** Diverse experimental baselines along with corpus analysis.

Time is not just metadata. It is a rhetorical device.



Thank you!

Uncovering Temporal Framing in the News



Project website + dataset

<https://mbzuai-nlp.github.io/temporal-framing/>

Presenter: **Tarek Mahmoud**

Appendix

Multi-label Temporal Framing Classification

You are an expert annotator for temporal framing in news text. Temporal framing refers to the rhetorical use of time-related elements to persuade, not merely to report chronology. A sentence should be labeled only if the temporal element contributes to its persuasive force.

Temporal framing taxonomy (short definitions and examples):

{BRIEF_TAXONOMY}

Key rules for annotation:

- Do not annotate factual chronological reporting or purely descriptive temporal mentions.
- Do not annotate quoted or indirect speech.
- If removing the temporal expression does not change the persuasive force, do not annotate.
- Multiple temporal frames may apply to a single sentence.

Context (optional):

{CONTEXT}

Target sentence:

{TARGET_SENTENCE}

Return the applicable temporal frames in structured JSON.

Figure 6: **Prompt template for temporal framing classification.** Taxonomy definitions from Figure 1b are inserted into the prompt via the BRIEF_TAXONOMY variable, after which the model predicts all applicable temporal frames in a single inference pass. The optional context block is included only in context-granularity ablation experiments.

Temporal Frame	Short Definition	Example
Primacy	<i>Significance attributed to being first in time.</i>	“The first to discover the cure will lead the world.”
Recency	<i>Significance attributed to the most recent events.</i>	“Today’s figures matter most for understanding public opinion.”
Urgency	<i>Emphasis on limited time or imminent consequences or threats.</i>	“Scientists warn the next 48 hours are critical.”
Temporal Anchoring	<i>Framing the discussion through the lens of past events.</i>	“We live in a post-9/11 world where security comes first.”
Nostalgia	<i>Invocation of a cherished past.</i>	“We must act now to bring back the good old days.”
Temporal Contrast	<i>Juxtaposition of “then” versus “now.”</i>	“Once a booming hub, now a city in decline.”
Continuity	<i>Persistence across time.</i>	“The economy has been climbing steadily for a decade thanks to policy changes.”
Skeptical	<i>Casting doubt about the future.</i>	“The government’s plan may collapse under financial pressure.”

(b) **Brief overview of the temporal framing taxonomy with definitions and examples.** For detailed definitions, additional examples, and rhetorical functions, see Appendix B.

Figure 1: An annotated excerpt and an overview of the temporal framing taxonomy.

Lang	Media Outlet	n	n_{opinion}	n_{sampled}
English	Forbes	397	271	53
	Breitbart	224	80	29
	The Guardian	385	119	28
	Daily Mail	882	108	27
	CNN	356	33	18
	Zero Hedge	104	70	17
	BBC	404	17	11
	Fox News	243	27	10
	China Daily (EN)	148	26	9
	CBS News	100	7	7
	SAMAA TV	57	5	6
	Daily Pakistan (EN)	42	5	6
	NBC News	109	6	5
	Time	28	10	3
	The Epoch Times	175	25	2
	Global Times (EN)	32	4	2
	Liberty Times Net (EN)	17	0	2
	Associated Press	10	2	1
	USA Today	2	1	1
	Kronen Zeitung (EN)	1	0	1
German	Focus Online	107	31	31
	t-online	242	18	28
	Die Welt	359	12	26
	n-tv	276	12	24
	Süddeutsche Zeitung	174	35	22
	Tages-Anzeiger	78	19	20
	Stern	87	12	15
	Bild	125	13	14
	Der Tagesspiegel	109	15	13
	NachDenkSeiten	13	13	11
	Kronen Zeitung	44	4	5
	Epoch Times (DE)	59	1	4
	Frankfurter Allgemeine Zeitung	70	28	3
	Junge Freiheit	5	4	3
	FAZ Kaufkompass	4	4	1

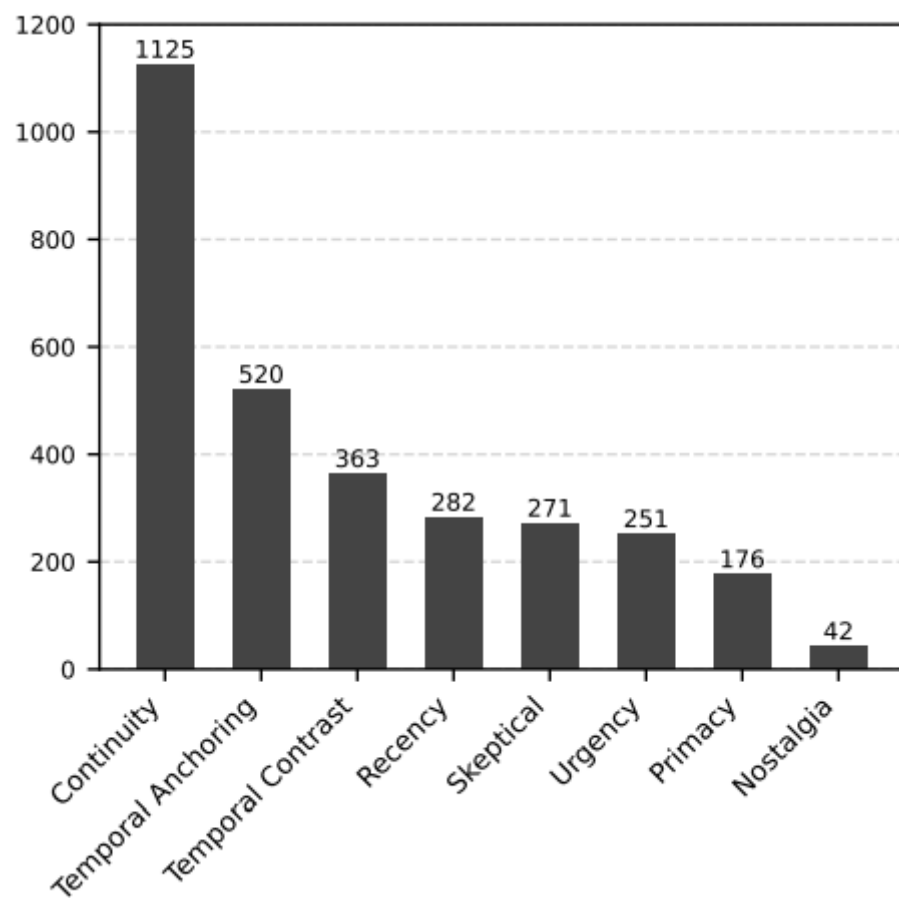
Table 1: **Media outlet-level coverage and final sampling counts.** n = successfully extracted articles. n_{opinion} denotes LLM-tagged indicators of opinionated content, used for sampling only. n_{sampled} denotes articles retained after stratified uniform sampling over time, with upsampling of opinionated articles.

Lang.	N	α_{det}	$\overline{\alpha}_{\text{frame}}$	min	max	Set-F1 ₊
EN	1,934	0.605	0.494	0.416	0.590	0.402
DE	2,317	0.720	0.588	0.302	0.768	0.443

Table 2: Sentence-level inter-annotator agreement. α_{det} denotes Krippendorff’s α (nominal) for binary detection of any temporal framing. $\overline{\alpha}_{\text{frame}}$ is the mean per-frame α across the eight temporal frames. Set-F1₊ is computed on the positive-only subset and averaged over the sentences.

Lang.	Docs	Sents	Words	Chars	AVG_s	AVG_w	AVG_c	SentAnn	Ann	AVG_s	AVG_a
EN	238	8,750	161,311	992,556	36.76	677.78	4,170.40	1,748	2,334	7.34	1.34
DE	220	11,909	163,381	1,147,357	54.13	742.64	5,215.26	617	696	2.80	1.13
Total	458	20,659	324,692	2,139,913	45.11	708.93	4,672.30	2,365	3,030	5.16	1.28

Table 3: **Corpus statistics.** The table shows the total number of documents (Docs), sentences (Sents), words (Words), and characters (Chars) by language. Averages per document are provided for sentences (AVG_s), words (AVG_w), and characters (AVG_c). The table also reports the number of sentences containing at least one annotation (SentAnn), the total number of annotations (Ann), the average number of annotated sentences per document (AVG_s), and the average number of annotations per annotated sentence (AVG_a).



Method	Model	Binary Detection			Multilabel Classification			
		P	R	F1	P	R	Micro-F1	Macro-F1
	Random Baseline	0.11	0.88	0.20	0.02	0.49	0.04	0.03
Zero-Shot	Llama-3.1-8B-Instruct	0.14	0.94	0.24	0.06	0.34	0.10	0.11
	Llama-3.3-70B-Instruct	0.28	0.78	0.41	0.16	0.43	0.23	0.21
	Qwen3-8B	0.21	0.75	0.33	0.08	0.29	0.13	0.14
	Qwen3-32B	0.36	0.57	0.44	0.20	0.31	0.25	0.22
	Qwen3-235B-A22B	0.31	0.70	0.44	0.17	0.38	0.24	0.23
	GPT-5.2	0.34	0.67	0.45	0.24	0.46	0.31	0.29
Supervised	XLM-R	0.38	0.75	0.51	0.28	0.57	0.37	0.33
	Llama-3.1-8B-Instruct	0.67	0.45	0.54	0.48	0.37	0.42	0.35
	Qwen3-8B	0.60	0.54	0.57	0.48	0.40	0.44	0.35

Table 4: **Overall results on the test set.** Zero-shot models show high recall but low precision, while supervised models achieve more balanced performance with strong gains in detection and multilabel classification. The random baseline samples labels from the annotation label distributions in the combine training and development sets.

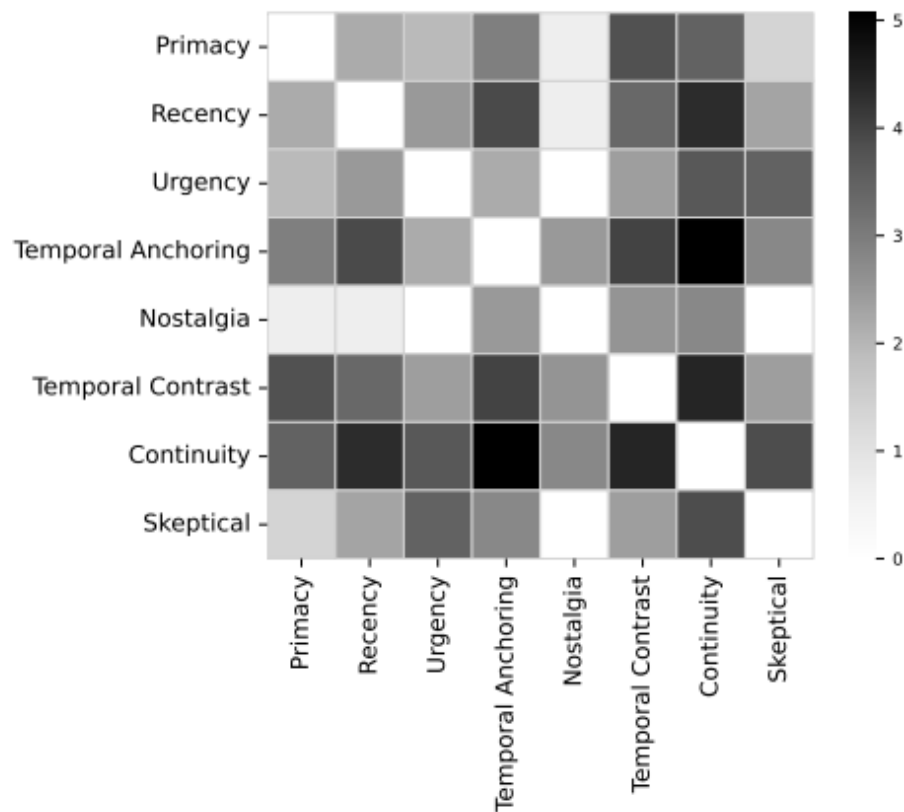


Figure 4: **Log-scaled heatmap of co-occurrence of temporal frames in the corpus.** Darker cells indicate more frequent co-occurrence patterns.

Temporal Frame	Most Associated Terms
Primacy	<i>record, arrivals, first, ever, democratic, lot, never, history, ufc, high, year, since</i>
Recency	<i>recently, latest, recent, navy, southern, hours, weeks, week, tariffs, friday, brought, study</i>
Urgency	<i>clinicians, essential, medical, must, soon, threat, rise, citizens, critical, needs, military, today</i>
Temporal Anchoring	<i>neocon, battle, copy, century, civil, refused, term, harry, pandemic, rural, reminder, remained</i>
Nostalgia	<i>you, have</i>
Temporal Contrast	<i>previous, longer, amendment, ago, plastic, now, once, become, today, until, research</i>
Continuity	<i>steady, increasingly, repeatedly, streak, spirit, methods, scandal, remained, cryptocurrency, continues, remains, volatile</i>
Skeptical	<i>non-citizens, spouses, affect, non-citizen, qdot, estate, laws, spouse, senators, planning, broader, tariffs</i>

Table 10: Top terms most strongly associated with each temporal frame, estimated using log-odds keyness. Words are shown in descending order of association strength.

	Train	Dev	Test	All
English	138/166 (83.1%)	27/33 (81.8%)	33/39 (84.6%)	238
German	95/153 (62.1%)	18/30 (60.0%)	24/37 (64.9%)	220
Total	233/319 (73.0%)	45/63 (71.4%)	57/76 (75.0%)	458

Table 9: **Corpus distribution across train, dev, and test splits.** Entries show *framed/total* documents per split (percentage in parentheses); for example, *138/166 (83.1%)* indicates that 138 of 166 English training articles contain at least one temporal framing instance.

Context Mode	Micro P	Micro R	Micro F1	Macro F1	Cost (\$)
Sentence-only	0.212	0.476	0.294	0.251	3.50
Prev + Target + Next	0.226	0.382	0.284	0.251	3.84
Full document	0.216	0.411	0.283	0.252	11.41

Table 11: Effect of context granularity on zero-shot temporal framing classification. Sentence-only input achieves the best performance while incurring the lowest cost, yielding the most efficient overall configuration.

Split	Model	Pri	Rec	Urg	Anch	Nos	Ctr	Cnt	Skp
EN + DE	<i>Zero-shot</i>								
	LLaMA-3.1-8B	0.11	0.20	0.06	0.13	0.10	0.14	0.16	0.00
	LLaMA-3.3-70B	0.16	0.17	0.24	0.21	0.19	0.23	0.33	0.17
	Qwen3-8B	0.03	0.10	0.16	0.15	0.11	0.23	0.27	0.07
	Qwen3-32B	0.19	0.28	0.18	0.26	0.16	0.23	0.31	0.18
	Qwen3-235B	0.16	0.22	0.21	0.24	0.32	0.26	0.30	0.12
	GPT-5.2	0.18	0.25	0.20	0.23	0.07	0.27	0.29	0.11
	<i>Supervised</i>								
	LLaMA-3.1-8B-Instruct	0.41	0.45	0.32	0.38	0.00	0.42	0.50	0.34
	Qwen3-8B	0.39	0.49	0.31	0.34	0.00	0.41	0.55	0.28
EN	<i>Zero-shot</i>								
	LLaMA-3.1-8B	0.16	0.28	0.12	0.25	0.19	0.20	0.22	0.00
	LLaMA-3.3-70B	0.20	0.25	0.30	0.28	0.30	0.42	0.40	0.17
	Qwen3-8B	0.08	0.21	0.25	0.25	0.26	0.30	0.35	0.05
	Qwen3-32B	0.28	0.44	0.25	0.34	0.19	0.29	0.32	0.10
	Qwen3-235B	0.26	0.36	0.29	0.37	0.38	0.36	0.36	0.05
	GPT-5.2	0.23	0.31	0.28	0.33	0.18	0.36	0.41	0.15
	<i>Supervised</i>								
	LLaMA-3.1-8B-Instruct	0.34	0.49	0.34	0.42	0.00	0.46	0.53	0.34
	Qwen3-8B	0.43	0.52	0.24	0.40	0.00	0.48	0.60	0.22
DE	<i>Zero-shot</i>								
	LLaMA-3.1-8B	0.06	0.06	0.02	0.04	0.06	0.00	0.05	0.00
	LLaMA-3.3-70B	0.09	0.06	0.14	0.14	0.13	0.08	0.20	0.18
	Qwen3-8B	0.01	0.03	0.08	0.07	0.04	0.13	0.16	0.08
	Qwen3-32B	0.10	0.09	0.09	0.17	0.12	0.10	0.28	0.29
	Qwen3-235B	0.06	0.07	0.13	0.13	0.27	0.16	0.18	0.23
	GPT-5.2	0.11	0.14	0.12	0.13	0.00	0.16	0.17	0.08
	<i>Supervised</i>								
	LLaMA-3.1-8B-Instruct	0.67	0.20	0.27	0.10	0.00	0.20	0.33	0.35
	Qwen3-8B	0.30	0.36	0.48	0.07	0.00	0.26	0.40	0.40

Table 12: **Per-label F1 scores across language evaluation splits.** Cnt = *Continuity*, Nos = *Nostalgia*, Pri = *Primacy*, Rec = *Recency*, Skp = *Skeptical*, Anch = *Temporal Anchoring*, Ctr = *Temporal Contrast*, Urg = *Urgency*.

Split	Model	Binary Detection			Multilabel Classification			
		P	R	F1	P	R	Micro-F1	Macro-F1
EN + DE	Random Baseline	0.11	0.88	0.20	0.02	0.49	0.04	0.03
	<i>Zero-shot models</i>							
	LLaMA-3.1-8B	0.14	0.94	0.24	0.06	0.34	0.10	0.11
	LLaMA-3.3-70B	0.28	0.78	0.41	0.16	0.43	0.23	0.21
	Qwen3-8B	0.21	0.75	0.33	0.08	0.29	0.13	0.14
	Qwen3-32B	0.36	0.57	0.44	0.20	0.31	0.25	0.22
	Qwen3-235B	0.31	0.70	0.44	0.17	0.38	0.24	0.23
	GPT-5.2	0.34	0.67	0.45	0.24	0.46	0.31	0.29
	<i>Supervised fine-tuning</i>							
	XLNet	0.38	0.75	0.51	0.28	0.57	0.37	0.33
	Llama-3.1-8B-Instruct	0.67	0.45	0.54	0.48	0.37	0.42	0.35
	Qwen3-8B	0.60	0.54	0.57	0.48	0.40	0.44	0.35
EN	Random Baseline	0.20	0.88	0.32	0.03	0.48	0.06	0.06
	<i>Zero-shot models</i>							
	LLaMA-3.1-8B	0.25	0.92	0.39	0.12	0.37	0.19	0.18
	LLaMA-3.3-70B	0.43	0.73	0.54	0.26	0.43	0.32	0.29
	Qwen3-8B	0.41	0.69	0.52	0.18	0.29	0.22	0.22
	Qwen3-32B	0.49	0.51	0.50	0.32	0.29	0.31	0.28
	Qwen3-235B	0.49	0.64	0.56	0.30	0.36	0.33	0.30
	GPT-5.2	0.46	0.58	0.52	0.37	0.44	0.40	0.26
	<i>Supervised fine-tuning</i>							
	LLaMA-3.1-8B	0.68	0.52	0.59	0.48	0.42	0.45	0.36
	Qwen3-8B	0.67	0.58	0.62	0.54	0.42	0.47	0.36
DE	Random Baseline	0.05	0.90	0.10	0.01	0.53	0.02	0.02
	<i>Zero-shot models</i>							
	LLaMA-3.1-8B	0.07	0.99	0.12	0.02	0.25	0.03	0.04
	LLaMA-3.3-70B	0.16	0.91	0.27	0.07	0.45	0.12	0.13
	Qwen3-8B	0.11	0.93	0.19	0.03	0.30	0.05	0.07
	Qwen3-32B	0.23	0.72	0.35	0.10	0.37	0.16	0.15
	Qwen3-235B	0.18	0.85	0.29	0.08	0.43	0.13	0.15
	GPT-5.2	0.21	0.83	0.33	0.14	0.64	0.23	0.17
	<i>Supervised fine-tuning</i>							
	LLaMA-3.1-8B	0.61	0.24	0.34	0.46	0.20	0.28	0.26
	Qwen3-8B	0.43	0.43	0.43	0.33	0.33	0.33	0.28

Table 13: Overall results broken down by language evaluation splits.

